P07

Modal Analysis and Orthogonality in Multi-Degree Systems

Authors D. J. Ewins

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Domain DOM_ENG

Abstract (quoted verbatim from the publication)

Multi-degree-of-freedom structures possess coupled mass and stiffness matrices. Generalized eigenvalue problems yield orthogonal mode shapes that diagonalize system matrices into decoupled modal oscillators.

Abstract quoted from D. J. Ewins et al., Modal Testing: Theory and Practice (1984).

Notes: record the page or section for every claim you make about this paper.

Reading prompts

1. What is the core problem addressed and what assumptions are made?
2. What mathematical or structural abstractions represent the physical phenomena?
3. How can this concept be mapped or transferred to analogous problems in other domains?

This card is a workshop wrapper prepared by the organisers. Only the abstract is quoted from the publication; read the paper itself through the link above before asserting any relationship.